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MIL-S-28683
25 July 1990

MILITARY SPECIFICATION

STOKERS, MECHANICAL: FOR PACKAGED BOILERS FIRED ON COAL OR A COMBINATION OF COAL-GAS, COAL-OIL, OR COAL-GAS-OIL (2,000,000 TO 50,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single mechanical stokers installed on packaged boilers fired by coal, coal and gas, coal and oil, or a combination of coal, oil and gas.

1.2 Classification The stoker will be of one the following styles as specified (see 6.2).

Style A - Spreader stoker with intermittent action grate.
Style B - Spreader stoker with traveling or chain grate.
Style C - Overfeed stoker with intermittent action grate.
Style D - Overfeed stoker with traveling or chain grate.
Style E - Underfeed stoker with retort grate.
Style F - Fixed grate stoker.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Huenexne, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4530

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2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents will be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-B-3180 - Boiler and Related Equipment, Packaging Of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following other documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Recommended Design Guidelines for Stoker Firing of Bituminous Coals.

(Application for copies should be addressed to the American Boiler Manufacturers Association, 950 N. Glebe Rd., Suite 160, Arlington, VA 22203.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code.
ASME PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM D 1857 - Fusibility of Coals.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies.
NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Standards for Motors and Generators

(Application for copies should be addressed to the National Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 851 - Guidelines for Stoker Operation.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The stoker shall either be furnished as part of a new boiler or shall be a replacement or conversion on an existing boiler as specified (see 6.1.1, 6.2, and 6.4). The stoker shall be of the style specified (see 1.2, 6.1.1 and 6.2).

3.2 Standards. The stoker shall conform to the applicable guidelines and requirements of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals and NFPA 851 Guidelines for stoker operation.

3.3 Standard commercial product . The stoker shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the stoker being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new.

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3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The stoker shall be designed in accordance with the guidelines and requirements of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals and any additional requirements specified here in.

3.6.1 Thermal input capacity. The thermal input capacity of the stoker shall be sufficient so as to allow the boiler in which the stoker is to be installed in to meet the boiler thermal output requirement in British thermal units per hour (Btu/hr) as specified (see 6.1.1 and 6.2).

3.6.2 Grate heat input rates. The stoker shall have a grate heat Input rate as required in 3.6.2.1 through 3.6.2.3 for the style specified.

3.6.2.1 Spreader stoker grate heat input rates. The grate heat input rate for style A stokers shall be not less than 200,000 Btu/hr per square foot and not greater than 700,000 Btu/hr per square foot. The grate heat input rate for style B stokers shall be not less than 200,000 Btu/hr per square foot and not greater than 700,000 Btu/hr per square foot.

3.6.2.2 Overfeed stoker grate heat input rates. The stoker grate heat input rate for the style C or style D stoker shall be not greater than 450,000 Btu/hr per square foot.

3.6.2.3 Underfeed stoker grate heat input rates. The stoker grate heat input rate for a style E stoker shall be not greater than 450,000 Btu/hr per square foot.

3.6.2.4 Fixed grate stoker heat input rates. The stoker grate heat input rate for a style F stoker shall be not greater than 360,000 Btu/hr per square foot.

3.6.3 Coal characteristics. The description of the coal and testing procedures used to establish the coal characteristics shall be in accordance with Appendix A of the ABMA Recommened Design Guidelines for Stoker Firing of Bituminous Coals. The stoker shall be designed for coal of the following characteristics as specified (see 6.1.1, 6.1.2, and 6.2):

a. Proximate analysis - as received (weight percent)

(1) Volatile matter
(2) Fixed carbon
(3) Ash
(4) Moisture (total)
(5) Equilibrium moisture

b. Higher heating value

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c. Ash analysis (percent by weight)

d. Ash fusion temperature in reducing and oxidizing environments

(1) Hemispherical (height equals one-half width)

e. Free swelling index

f. Coal source

(1) State
(2) Mine

3.6.4 Coal size distribution. The size distribution of the coal provided to the hopper shall be within the size distribution recommendations of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals for the required stoker style.

3.6.5 Spreader stokers. Style A and style B stokers shall be furnished in accordance with the design requirements specified in 3.6.5.1 through 3.6.5.2.

3.6.5.1 Feeder. The spreader stoker shall be designed to burn a portion of the coal in suspension and the balance of the coal on the grates. The combined width of the coal feeder distributors shall be not less than 40 percent of the width of the stoker grate. Multiple feeders and distributors shall be able to be operated simultaneously or independently from the others. Any one feeder may be taken out of service without affecting the operation of the remaining feeders. The feeder system shall be able to evenly distribute the fuel over the grate surface. The fuel supply system shall be furnished with a forced lubrication system to bearings and other points of high wear. Each feeder shall have a separate manual speed bias control.

3.6.5.2 Grates. Grates shall allow an even distribution of air flow from the air plenum beneath the grate. The vertical fronts of this enclosure are to be fitted with cast-iron doors for each feeder and distributor.

3.6.6 Overfeed stokers Style C and style D stokers shall be furnished in accordance with the design requirements specified in 3.6.6.1 through 3.6.6.3.

3.6.6.1 Intermittent action grate. The style C intermittent action grate drive mechanism shall provide l positive motion to the grates to ensure proper movement of the ash bed toward the ash pit and to feed the coal from the hopper. Operation of the grate drive shall be intermittent with the frequency and duration of operation controlled manually or by adjustable time clocks. All motors, shafts, bearings, and other equipment shall be installed outside the stoker and boiler enclosures for ready access for servicing and maintenance without requiring a unit shutdown.

3.6.6.2 Traveling or chain grate. Style D traveling or chain grate shall have links with uniform length and cross section to provide uniform air spacing and air distribution. No parts of the grate shall extend into the

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fire. The grate shall be designed to allow repair links to be installed without taking the stoker out of service. Provisions shall be made for adjusting the tension in the chain grate from the front of the boiler without requiring the stoker to be stopped.

3.6.6.3 Feed gate. The feed gate shall have removable and replaceable cast-iron shoes and refractory lining. The feed gate shall be readily adjustable with an indicator provided to show the depth of the fuel bed.

3.6.7 Underfeed stokers. Style E stokers shall be furnished in accordance with the design requirements specified in 3.6.7.1 through 3.6.7.3.

3.6.7.1 Coal feed device. Each stoker shall have a center retort that has coal pushed into it from the coal hopper by a ram, plunger, wormfeed screw or other device. Distribution and agitation in the retort and fuel bed shall be provided by the coal feed device. The stoker design shall provide that a ram or plunger is always on the full stroke regardless of the rate of feed to ensure agitation of the fuel bed.

3.6.7.2 Retort grate. The grate surface shall consist of plate type tuyeres and shall have the proper width and length to support the boiler capacity requirements. The grate dumping mechanism shall be furnished with the necessary gears to allow smooth manual operation.

3.6.7.3 Ash quench piping. The stoker shall be provided with manually operated perforated water spray pipes in the undergrate ash collection area.

3.6.8 Fixed grate stokers. Style F stokers shall be furnished in accordance with the design requirements specified in 3.6.8.1 through 3.6.8.3.

3.6.8.1 Coal Feed. The stoker shall have either a center retort or a simple overhead spreader coal feed mechanism. The center retort shall have coal pushed into it from the coal hopper or bin by a ram, plunger, worm feed screw or other device. The ram or plunger feed mechanism shall always use a full stroke regardless of feed rate. An overhead spreader shall evenly distribute the coal over the entire bed area.

3.6.8.2 Fixed grate. The grate shall distribute combustion air evenly across the bed. Tuyeres shall be supplied with center retort feed nechanisms. Ash shall be able to fall through the grate to an ash collection bin. The grate shall be of the dead plate or refractory type stationary hearth.

3.6.8.3 Control system. The automatic air control and coal feed systems shall either be on-off, high-low-off, or modulating, as specified (see 6.2).

3.6.9 Stoker turndownw. The turndown of the stoker shall be made during the normal firing cycle without interruption or change of stoker components. Stable stoker operation shall exist at the minimum firing rate. Style A and style B spreader stokers shall have a turndown ratio of 3 to 1. Style C and style D overfeed stokers shall have a turndown ratio of 5 to 1. Style E underfeed stokers shall have a turndown ratio of 8 to 1.

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3.7 Performance. The thermal performance of the stoker shall be based upon the ability of the boiler to produce the required thermal output when the stoker is fired using the coal described in 3.6.4. The performance of the boiler shall be evaluated in accordance with the ASME PTC 4.1 abbreviated efficiency test using the heat loss method. Operation of the stoker equipment shall be performed to the requirements of this specification (see 6.1.1).

3.8 Construction. The stoker shall be constructed to meet the design and performance requirements specified herein. The stoker shall be furnished with all necessary shafting, couplings, bearings, drives with guards and safety devices, linkage, rods, and other equipment that are required for safe operation of the stoker. Safety release devices such as shear pins or stall clutches shall be provided to prevent damage to the stoker due to foreign materials in the coal. All necessary structural supports above the floor level shall also be provided.

3.8.1 Stoker doors. The front wall of the stoker shall be steel or cast-iron reinforced with steel to suit the type of stoker application. The front wall shall be of the proper size to match the boiler and bolt to the boiler to form a gas-tight seal. The front of the stoker shall be provided with refractory lined fire doors with safety latches. Necessary furnace and other access doors required for safe, continuous operation of the stoker and boiler shall be provided. The hopper, fan, air duct, and other external equipment shall be located so as to not obstruct access to doors. Style F units are not required to be furnished-with stoker doors and fronts.

3.8.2 Hopper. Each coal feeder shall be furnished with a rigidly reinforced coal hopper. The capacity of each hopper shall be as specified (see 6.1.1 and 6.2). A cleanout door shall be provided on the hopper to allow for removal of foreign objects that may become lodged in the hopper.

3.8.3 Grate temperature indicators. When specified (see 6.2), grate temperature indicators shall be furnished. The number of grate temperature indicators shall be as specified (see 6.2). Temperature indicators shall be placed at the grate positions where highest temperatures are expected, The dial or digital temperature display shall be located in the control panel. A common display may be used to indicate the temperature readings with a switch to allow each indicator to be read on the display.

3.8.4 Overfire air system. A high pressure overfire air system with a manual override shall be furnished with the stoker. The system will include a high pressure, electric motor driven, overfire air fan. The equivalent capacity of available overfire air shall be as specified in the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals for the required stoker style. Sufficient overfire air jets shall be provided and properly located to create the necessary furnace turbulence when operated in conjunction with the cinder return jets if provided. The overfire air system shall be able to be connected to the fuel feed regulating system so that the overfire air volume and fuel rates are synchronized automatically. A manual adjustment shall be provided to regulate the ratio of the overfire air to the fuel feed rate. A manual override of the automatic system shall also be provided.

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3.8.5 Cinder return system. When specified (see 6.1.1 and 6.2), a cinder return system shall be furnished. The cinder return air flow is not required to be modulated with fuel feed rates. Cinder return piping shall be arranged with a minimum of bends. Piping shall be heavy wall pipe with hardened bends.

3.8.6 Blast gates. The stoker shall be provided with manually operated stoker blast gates and operating handles. The blast gates shall be installed at the stoker air plenum inlet. Style F stokers may be furnished without blast gates.

3.8.7 Primary air supply system. The primary air supply equipment shall be sized and arranged to ensure a continuous steady air flow adequate for all operating conditions of the stoker. The forced draft primary air fan shall discharge air directly to the stoker air plenum. A variable fan damper shall be provided that can be operated simultaneously with the coal feeding mechanism to proportion the air flow to the coal feed rate. Style F stokers shall be furnished with an on-off, hi-low-off, or modulating control mechanism (see 3.6.8.3). The arrangement of air inlets and ductwork shall minimize contamination of the air supply by such materials as water and fuel. Manually controlled individual air zones shall be provided to allow for distribution of underrate air on style C and style D overfeed stokers. Drains and access openings shall be provided. The underside of the grates are to be enclosed so as to form a plenum chamber. An effective air seal shall be provided on the sides of the grate that are not discharging ash. The end of the grates; where the ash is discharged into bunker or pit shall be enclosed with a dust tight enclosure made up of heavy metal plates, properly protected with refractory where they are exposed to the furnace. This chamber is to be provided with hoppers, l ccess doors, and supporting steel. Metal or concrete air ducts from the forced draft fan to the plenum chamber shall be provided.

3.8.8 Structural supports. All structural supports above the operating floor line shall be provided for the stoker, stoker drives, and coal hoppers. The supports shall be so located and protected that they will not be overheated or damaged from the heat of the combustion either from the furnace or collected material in the ash pit.

3.9 Combustion control system. Adjustable linkages shall be provided to modulate the fuel, air input, and draft damper in accordance with boiler load demand on stoker fired boilers. Style F stokers shall be furnished with an on-off, hi-low-off, or modulating control mechanism (see 3.6.8.3). The relative rates of change shall be controlled simultaneously so as to maintain the air to fuel ratio within the limits required for continuous combustion. Means shall be provided to prevent the control system from demanding fuel-rich mixture during normal operation or start-up, Control action toward more fuel and less air shall be blocked when the air to fuel ratio is decreased below a preset limit. This block shall be in effect with either automatic or manual control.

3.9.1 Grate speed or ash discharge rate adjustment. When specified (see 6.2.), a linkage shall be provided on style A or style B spreader stokers on the fuel feed regulating system so that the grate speed and the fuel feed

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rates are synchronized automatically. A manual adjustment shall be provided to regulate the ratio of the grate speed or ash discharge rate to the fuel feed rate in order to compensate for variations in the ash content of the fuel. A manual override of the automatic system shall also be provided to allow for the manual adjustment of the grate speed or ash discharge rate.

3.9.2 Control system sygnal type. The control system shall use electric, pneumatic, or hydraulic signals as specified (see 6.2).

3.10 Electrical requirements. Electrical wiring design shall be in accordance with the applicable requirements of NFPA 70. The stoker and associated equipment shall obtain power from a circuit with a single manual service disconnect switch and automatic circuit breaker for all boiler electrical equipment. Motors shall be in accordance with the provisions of NEMA MG 1, shall be totally enclosed fan cooled, and designed for operation on the specified voltage, phase, and frequency (see 6.2). Unless otherwise specified (see 6.2), motors of 100 horsepower and less shall be provided with magnetic starters and overload protection. Motors over 100 horsepower shall have reduced voltage starting. Motor starters shall conform to the requirements of NEMA ICS 1 and ICS 2. Motor starters and controls shall be enclosed in NEMA ICS 6 type 12 enclosures.

3.11 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or other contaminating substances. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer’s current materials according to manufacturer’s current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.12 Identification marking. Identification shall be permanently and legibly marked directly on the stoker or on a corrosion-resistant metal plate. Identification shall include manufacturer's name, stoker model and serial number, and year of-manufacture. The Department of Defense (DoD) contract number that this equipment is procured under shall also be marked on the equipment or identification plate.

3.13 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each stoker. When furnished, the spare parts required and the quantity thereof shall be as specified (see 6.2).

3.14 Firing tools. A complete set of manufacturer’s firing tools shall be furnished with the stoker including a slice bar, clinker hook, and ash hoe.

3.15 Factory start-up service. Unless otherwise specified (see 6.2), factory service personnel shall be provided to supervise installation and start-up of the stoker for five consecutive 8-hour working days.

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3.16 Workmanship. Workmanship shall be of such quality as to produce stokers that meet the requirements of this specification and standards prevailing among manufacturers who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascetiain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1 Responsiblity for inspecection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance inspection. This inspection shall be performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the quality conformance inspection shall include the examination of 4.3. The packaging inspection of 4.5 is required only when direct shipment to the installation site is expected.

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4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), on-site inspection shall consist of the tests of 4.4.

4.3 Examination. Each stoker and related equipment supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability.

4.4 On-site tests. Operational fire tests and equipment tests shall be performed on each installed stoker.

4.4.1 Operational fire tests. Each stoker shall demonstrate control and operational conformance to the requirements of this specification under varying load conditions that range from the maximum required capacity to the minimum required capacity without on-off cycling. Stoker capacity shall be adequate if the new or clean boiler it is installed in is able to meet the maximum thermal capacity and efficiency requirements. Stokers that exhibit (1) excessive smoking, (2) nuisance shutdown due to faulty stoker control operation, (3) uneven firing of the coal bed, (4) inconsistent bed movement and ash dumping equipment, or (5) necessity for frequent or difficult adjustments shall be rejected. All safety devices and trips shall be tested prior to the operational tests. Operational tests shall be as specified in 4.4.1.1 through 4.4.1.5.

4.4.1.1 Air infiltration. The unit shall be tested for excessive air infiltration by using a smoke bomb or other method.

4.4.1.2 Stoker turndown ratio. The specified stoker turndown ratio (see 3.6.9), shall be verified by firing the unit at the minimum continuous firing rate required.

4.4.1.3 Combustion controls. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the load demand though the entire firing range required by the stoker turndown ratio. The transfer of control from automatic to manual and from manual to automatic shall be performed on the overall system and independently on each automatic/ manual subsystem. This transfer shall be smooth without the need for special adjustments to trim the combustion control devices. The air/fuel ratio controller shall be tested by attempting to create a fuel rich operating condition below the manufacturer's pre-set limit.

4.4.1.4 Ash system. The ash system shall be demonstrated that it can be operated without plugging under normal operating conditions.

4.4.1.5 Ash piling. Operate soot blowers (under normal operating conditions) with the boiler in automatic. Observe that extreme upset condition and ash piling on the grate do not occur.

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4.4.2 Equipment tests. Linkages, grate movement mechanisms, fans, feed hoppers, feed distribution systems, ash quench spray systems, draft regulators, electric motors and other stoker related equipment shall be operated as part of the operational fire tests. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of mechanical devices shall be smooth without backlash.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180.

5. PAGKAGING

5.1 Preservation. packing. and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3180 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The stokers covered by this specification are intended for use In packaged boilers or hot water generators fired by coal, coal and oil, coal and gas, or a combination-of coal, oil, and gas.

6.1,1 Selection factors. The following items should be considered in the selection of a stoker for boilers fired on coal and the exercise of options here in:

a. Styles A and B spreader stoker styles are the most responsive to load changes and can handle most of the available types of coal. It is recommended to use this type of stoker whenever a turndown ratio of not less than 33 percent is acceptable. Spreader stokers are recommended if low Btu western coal is to be burned in the boiler (see 3.1).

b. Styles A and C intermittent action stokers include vibrating, pulsating, reciprocating and other noncontinuous grate movement mechanisms (see 3.1).

c. Styles C and D overfeed stokers may require special arch design if fired on lower volatile coals or high moisture, low Btu western coal. The characteristics of coal from probable sources other than that described in the specification should be considered to insure compatibility with the stoker design. It may be necessary to provide more than one coal specification. Styles C and D overfeed stokers are not recommended when the free swelling index of the coal is over 5 (see 3.1).

d. Style E underfeed stokers are not recommended for boiler output capacity requirements greater than 25,000,000 Btu/hr or when the free swelling index of the coal is over 5 (see 3.1).

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e. Style F stokers are for small boilers between 2,000,000 and 7,000,000 Btu/hr. These stokers are not designed to burn low Btu western coal. Feed mechanisms can include a central retort or an overhead central gravity spreader. This style requires much less auxiliary hardware and is reconmended as an economical choice for small boilers.

f. Dumping grates are not recommended because of air pollution control restrictions and are typically used on boilers that cannot economically justify additional flue gas cleaning equipment (see 3.1).

g. This specification can be used for replacement or conversion installations to specify the procurement requirements for the stoker. However, it will remain the responsibility of the contractor who is performing the boiler modifications to make the retrofitted boiler operate satisfactorily (see 3.1).

h. The minimum heat duty for a stoker should be not less than 2,000,000 Btu/hr boiler output capacity because packaged boilers fired on coal are not economic when compared to those fired on liquid or gas fuel in this size range (see 3.6.1).

i. The maximum heat duty for a packaged stoker should be not greater than 50,000,000 Btu/hr boiler output capacity due to transportation restrictions for the assembled unit. larger stoker fired boilers should be field erected and are outside the scope of this specification (see 3.6.1).

j. Property ranges or values should not be used in the description of the coal characteristics (see 3.6.3).

k. The stoker can only be tested when fired as part of the boiler because the thermal output capacity of the boiler is the only performance test that relates to the performance requirement of the stoker (see 3.7).

l. Hopper capacity is dependent on method of coal supply to hopper. Spreader style stokers have a hopper capacity of about 200 pounds if a continuous feed system to the hopper is available (see 3.8.2).

m. Cinder return systems should be specified on all new packaged stoker fired boilers. It may be too difficult to economically install piping with a minimum of bends on replacement stokers (see 3.8.5).

n. Fungus resistant varnish, conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the boiler is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

o. Electromagnetic interference suppression (EMI), when required, the unit should meet the UM05 requirements for class C3, group III equipment as specified in MIL-STD-461. The stroker should be subjected to tests to determine conformance in accordance with MIL-STD-462.

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6.1.2 Coal and stoker selection criteria. The performance requirements specified in 3.7 can be met with other coal fuels providing the proper selection of the stoker, grates, coal sizing, and ash softening temperatures are made. A quantitative proximate analysis for coal and a quantitative ultimate analysis for oil or gas, depending on the fuels to be fired, should be made available to the contractor. Corrections between the values of the analysis in this specification and the values of the guaranteed analysis of the fuels to be fired, both at the time of the contract and at the time of testing, should be made in accordance with section 7 of ASME FTC-4.1. Further requirements for the purchase and analysis specifications of coal can be found in the ASTM standards. The stoker manufacturers have made certain recommendations on" coal specifications for various types of stokers and grates. The following coal analysis should be used as a guide only:

a. Coal (proximate analysis) for spreader stokers (Styles A and B)

(1) Moisture:	0 - 10	percent
(2) Volatile Matter:	30 - 40	percent
(3) Fixed Carbon:	40 - 50	percent
(4) Ash:	5 - 15	percent
(5) Higher heating value:	12,500	Btu/lb minimum (as fired basis)

Ash softening temperature: Not less than 2300°F as determined (where height of cone equals one-half width and where there is not greater than 20 percent Fe_2O_3 in ash) in accordance with ASTM D 1857.

b. Coal (proximate analysis) for overfeed stokers (Styles C and D):

(1) Moisture:	0 - 20	percent
(2) Volatile Matter:	30 - 40	percent
(3) Fixed Carbon:	20 - 64	percent
(4) Ash:	6 - 20	percent
(5) Free Swelling Index (FSI):	5	maximum
(6) Higher heating value:	10,500	Btu/lb minimum (as fired basis)

Ash softening temperature: Not less than 2100°F as determined (where height of cone equals one-half width and where there is not greater than 20 percent Fe_2O_3 in ash) in accordance with ASTM D 1857.

c. Coal (proximate analysis) for underfeed stokers (Style E):

(1) Moisture:	0 - 10	percent
(2) Volatile Matter:	30 - 40	percent
(3) Fixed Carbon:	40 - SO	percent
(4) Ash:	5 - 10	percent
(5) Higher heating value:	12,500	Btu/lb minimum (as fired basis)
(6) Free Swelling Index (FSI)	5	Maximum

Ash softening temperature: Not less than 2500°F as determined (where height of cone equals one-half width and where there is not greater than 20 percent Fe_2O_3 in ash) in accordance with ASTM D 1857.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style required (see 1.2 and 3.1).
c. Issues of DODISS cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. If stoker is part of a new packaged boiler or a replacement on an existing packaged boiler (see 3.1).
e. Boiler thermal output requirements (see 3.6.1).
f. Coal characteristics (see 3.6.3).
g. Automatic air control and coal feed system for style F stokers (see 3.6.8.3).
h. Capacity of hopper (see 3.8.2).
i. When grate temperature indicators are required, and if required, number of grate temperature indicators to be furnished (see 3.8.3).
j. When a cinder return system is to be furnished (see 3.8.5).
k. When Style A and Style B stokers are to be provided with a linkage for automatic synchronization of grate speed and fuel feed rates (see 3.9.1).
1. Signal type for the control system on style F boilers (see 3.9.2).
m. Voltage, phase, and frequency that motors are to be designed for (see 3.10).
n. When motors of 100 horsepower and less are not to be provided with magnetic starters and overload protection (see 3.10).
o. Color of finish coat (see 3.11).
p. When spare parts are required (see 3.13).
q. If required, description of spare parts and quantity thereof to be furnished (see 3.13).
r. When other than 5 consecutive 8 hour days of factory service personnel are to be provided to supervise installation and start-up of the stoker (see 3.15).
s. Quality conformance inspection, if other than as specified (see 4.2.1).
t. Tests for on-site inspection, if other than as specified (see 4.2.2).
u. Level of presentation and level of packing required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data item description (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/ provided and that the DID is tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (CDRL) (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 24.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
6.3.2, 6.3.3, 6.3.5, 6.3.1.1 and 6.3.l.2	MISC-80678	Certificate Data/Report
6.3.4	DI-M-2050	Commercial Manuals

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The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Compliance. Prior to approval of shipment, the contractor should submit to the contracting officer or authorized representative satisfactory evidence that the proposed stoker to be furnished under this specification meets the requirements of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals and NFPA 851 Guidelines for stoker operation.

6.3.1.1 ABMA requirements. Acceptable evidence of meeting the requirements of the ABMA Recommended Design Guidelines for Stoker Firing of Bituminous Coals should all be the manufacturer's certification of compliance with the requirements of these guidelines, as applicable (see 6.2.2).

6.3.1.2 NFPA 851 requirements. Acceptable evidence of meeting the requirements of NFPA 851 Guidelines for stoker operation should be the manufacturer's certification of compliance with the requirements of these guidelines, as applicable.

6.3.2 Design verification. The manufacturer should furnish data to verify compliance with the specifiedgrate heat input rate (See 3.6.2).

6.3.3 On-site tests. Operational fine tests and equipment tests should be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the stoker manufacturer is accepted at the discretion of the contracting officer or his authorized representative. Acceptable fire test results from identical or similar stoker equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured with this specification is in compliance with one or more of the tests of 4.4.1 through 4.4.2.

6.3.4 Technical manuals. The manufacturer should provide two commercial technical manuals normally prepared and supplied with the boiler.

6.3.5 Air/fuel ratio curve. The contractor should submit predicted air\fuel ratio curve for approval prior to startup of the boiler.

6.4 Definitions. Definitions used in this specification have been in accordance with the ABAA Lexicom of Boiler and Auxiliary Equipment. Other terms used in this specification should be interpreted in accordance with the referenced national standards.

6.4.1 Replacement. A stoker replacement is defined in this specification as the simple replacement of existing equipment without changing the fuel, thermal heat duty, or air pollution emissions.

6.4.2 Conversion. A stoker conversion is defined in this specification as the replacement of existing equipment to change the type of fuel used, the thermal heat duty, the boiler air pollution emissions or other significant operating variable.

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6.5 Subject term {key word) listing.

Ash removal
Cinder return system
Grate
Hopper
Overfeed stoker
Overfire air
Stoker
Stoker driver
Spreader stoker
Underfeed stoker

Custodians:
Army - ME
Navy - YD

Review Activity:
Army - CE

User Activity:
Navy - MC

Preparing Activity:
Navy - YD

(Project 4530-0040)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-28683	25 July 1990

3. DOCUMENT TITLE MILITARY SPECIFICATION - STOKERS, MECHANICAL: FOR PACKAGED BOILERS FIRES ON OR A COMBINATION OF COAL-GAS, COAL-OIL, OR COAL-GAS-OIL (2,000,000 TO 50,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME D. C. MUI, Project Manager Facilities Criteria Development Division Civil Engineer Support Office (CESO) Code 156	b. TELEPHONE (Include Area Code) (1) Commercial: (805) 982-5801 ext. 424 (2) AUTOVON: 551-5801 ext. 424
c. ADDRESS (Include Zip Code) Naval Construction Battalion Center CESO (Code 156) Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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